



Assignment 2 Specification

Networking using Connected Devices

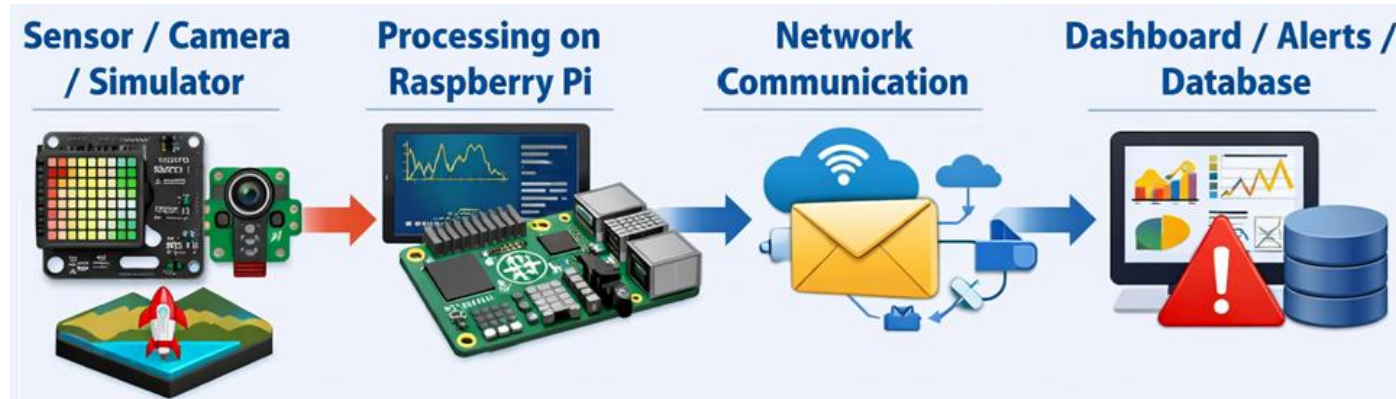
Produced by: Frank Walsh

Computer Systems and Networks

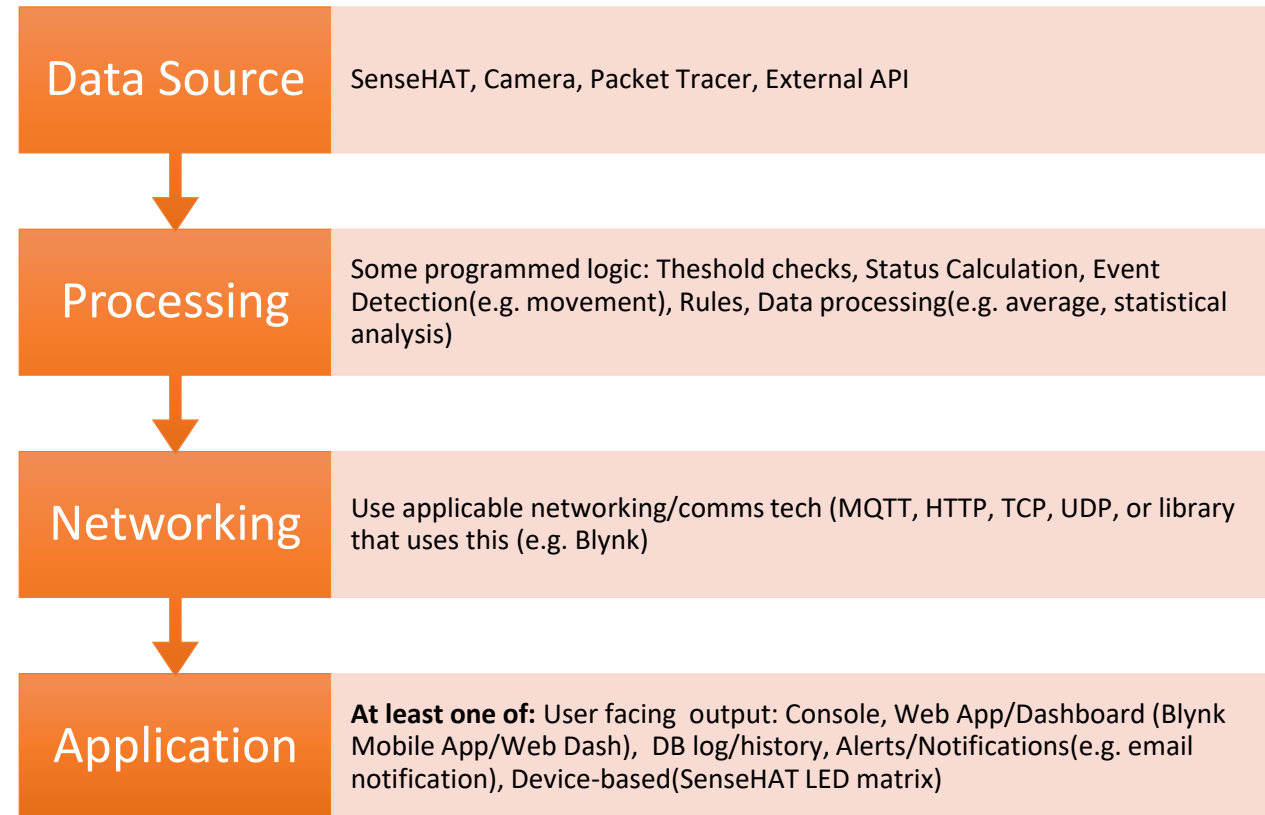
*Certificate/HDip Computer Science
2026*

Context

- 40% of your overall mark
 - Project ethos (1 overall project)
- For this assignment you will design and build a small connected device project:
 - Propose a project.
 - Create a working project.
 - Typically includes a "physical" aspect – sensors/devices(Rpi/SenseHat)
 - Present and communicate your work in a clear manner.



Requirements: Include the different layers (Sensors, processing, Network, Application) in a connected Project.



Proposed Timeline(5 Weeks)

April 26th

Proposal document

- Doesn't need to be exact/long!
- Sections:
 - Introduction
 - Proposed technologies (protocols/devices/prog lang)
 - Proposed Tools (IDEs, cloud platforms)
- **Deliverable: One page pdf or markdown document with general concept.**

May 29th

Presentation and interview:

- Git Repository submission(all code/resources)
- Include a **max 5 min** video demoing your project. (YouTube video)
- 10 min Zoom Interview
- **Deliverable: Accessible Github repo. Link to video.**

Proposal Document



Network and IOT Application.

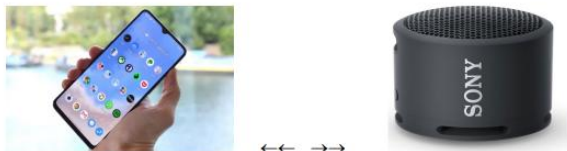
Git Repo:

[https://github.com/\[redacted\]_Application_Assignment](https://github.com/[redacted]_Application_Assignment)

I would like to restrict the time my son spends on his PS4. My proposal is as follows:



Using the Mac Address / IP Address of the PS4 which is on my home network when powered on, the Raspberry pi will detect when the PS4 is powered on, notify me through my web app to my mobile phone, start a countdown timer to 60 minutes. The sense hat will count down (in green) the minutes using the led's and when that 60 minutes has elapsed, the sense hat lights will turn red and tell the user via Bluetooth speaker to vacate the vicinity. The camera / motion sensor will be set so that after 2 minutes, if there is someone in the room, I will receive a message stating STILL GAMING!. If there is no movement, I will get a message saying ALL CLEAR.



PROJECT NAME SHOWER SAVE

10/11/2021

STUDENT NAME: [redacted] STUDENT ID: [redacted]

1. Project Background and Description

i The shower save system is a shower length measurement and alert system. This system has two possible purposes:

Use 1 - The purpose of this device is to notify someone by text and alarm that a person has been in the shower longer than expected. This device is useful for the guardians of anyone elderly or infirm to notify them by text or alarm, if the user has fallen in the shower and not exited.

Use 2 - This shower alarm could be used to measure the length of time spent in a shower in a country where water restrictions are in place.

Shower Save also measures shower lengths and records these times on a dashboard.

1. Raspberry Pi is placed on the outside of the shower door.
2. When the door opens the raspberry pi sends an event 1 which also contains temperature value (cold) - Person enters shower
3. When the door opens next time, the raspberry pi sends an event 2 with temperature value(hot) - Person exits the shower
4. The length of time between shower start events (door opens cold temp) and shower end events (door open hot temp) is measured over a period of time and saved to dashboard on phone app*
5. Write code to calculate the length of time between event 1 (cold) and event 2 (hot). If it is 5 minutes longer than average shower time, send text message to guardian/homeowner and trigger alarm.

*How will the raspberry pi know that it is the start of the shower versus the end of a shower. Only measure between an event 1 where the temperature is lower (Shower not on yet) and event 2 where the temperature is higher (shower is complete).

Notifications sent in the following scenario:

If Event 2 (high Temp) is longer than the average shower time (Time between events with low temp to events with high temps)

Alarm (either Phone App or Alexa)

If Event 2 (high Temp) is longer than the average shower time (Time between events with low temp to events with high temps)

Git and GitHub

- Git and GitHub are mandatory for this assignment.
- Your repo should:
 - Have regular commits
 - Clear commit messages
 - All your resources for the project
 - Source code
 - Documentation(e.g. proposal doc)
 - A README.md
 - Setup notes/supporting resources.

Scope

This is a **5-week project**, keep things manageable:

- one Raspberry Pi is enough
- simulated data is acceptable where appropriate
- One network protocol is enough
- Simple dashboard is enough
- Cloud/server deployment is optional – **not required**

You are not expected to build a production ready system. Think of it as a proof of concept demonstration.

Final Project Submission

chamberfoodsPublic

Watch1

[Code](#) [Issues](#) [Pull requests](#) [Actions](#) [Projects](#) [Security](#) [Insights](#)

master1 branch0 tags

[Go to file](#) [Add file](#) [Code](#)

chamberpic.html989c95b on Dec 20, 202143 commits

ChamberFoods	Commit changes 18/12/2021	11 months ago
assets	Add files via upload	11 months ago
website	Update chamberpic.html	11 months ago
README.md	Update README.md	11 months ago

README.md

Project Overview

[Chamber Foods](#) is an IoT project combining smart appliances, artisanal foods, and DIY.



About

No description, website, or topics provided.

[Readme](#)

0 stars

1 watching

0 forks

Releases

No releases published

Packages

No packages published

Languages

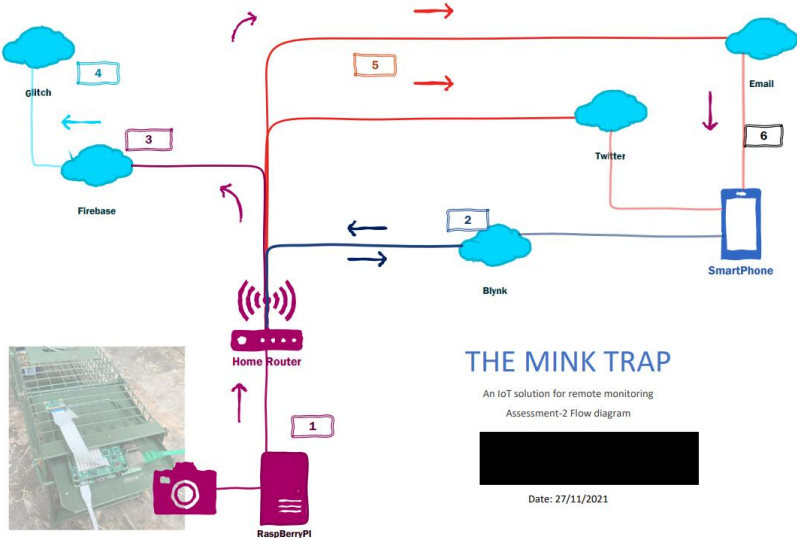
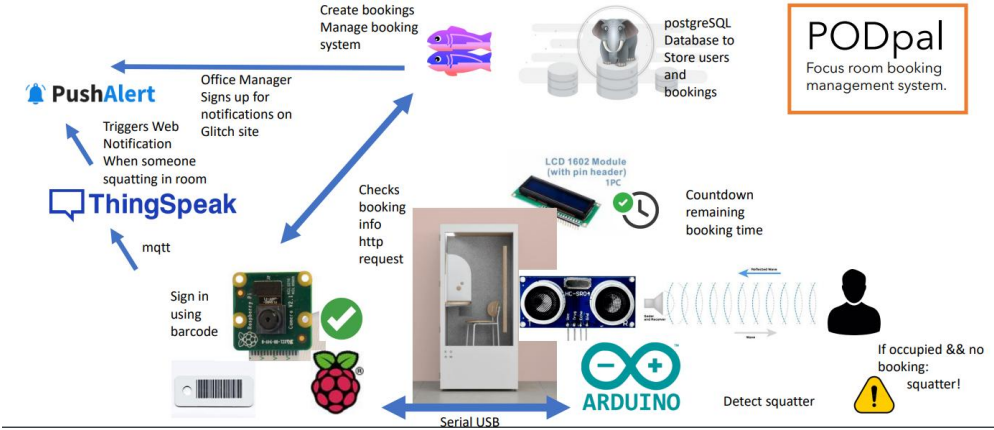
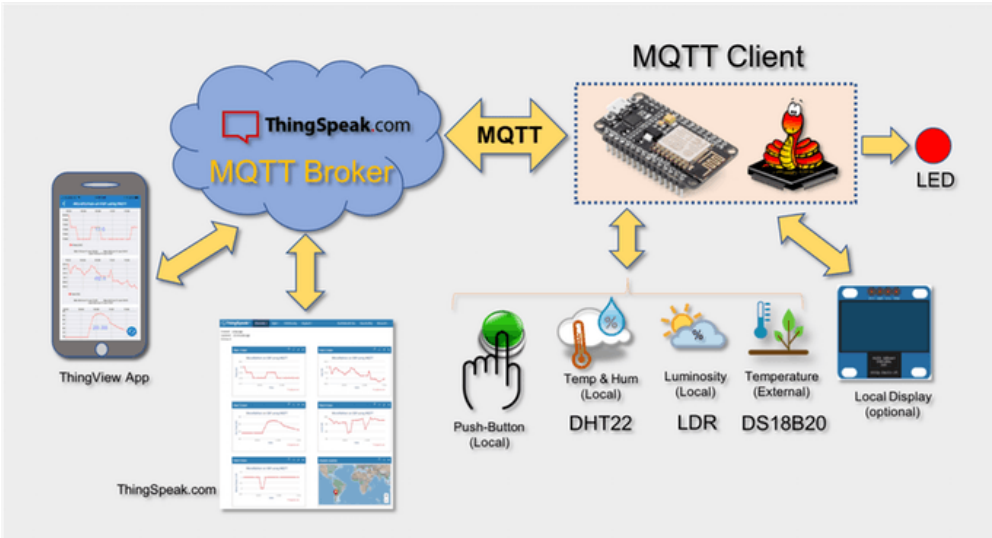
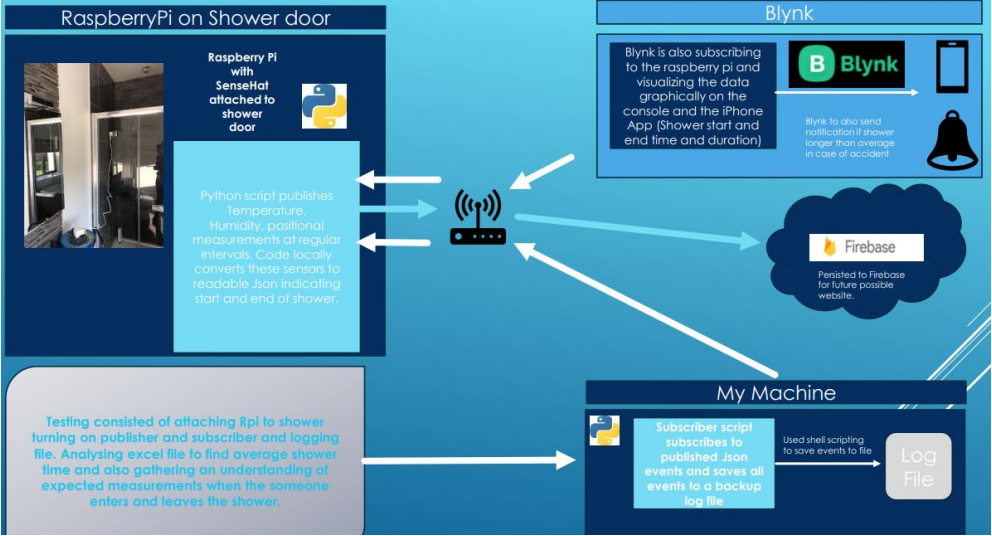
Python69.4%

C22.0%

HTML7.4%

Other1.2%

Project Graphic



Proposed Timeline

Week1

- Project idea, proposal, and approval

Week2

- Get input/data source working on the Raspberry Pi

Week3

- Implement networking/connection with other component/platform/service

Week4

- Add output layer such as dashboard, logging, alerts, or storage

Week5

- Testing, refinement, documentation, GitHub cleanup, and demo preparation



Grade Spectrum Guide

	IoT Solution	Networking/IoT Technologies Used	Combined knowledge	Communication
Release 1 Base (30-49)	One input source to Device	Physical/Data link layer solution. Basic one way connection between device/processes.	2 programme strands present in output. Basic knowledge of each exhibited. (e.g. programming, database, computer systems)	Zip file and/or basic Repo: Minimal (1) communication resource used (e.g. simple readme.md) and video.
Release 2 Good (50-64)	At least one real or simulated input source	Wireless/Wired protocols including network and transport layer. Interconnected device(s) and or processes.	Apply and combine concepts from more than two modules/strands..	Good Github Rep: Repository includes clear structure, documentation.
Release 3 Excellent (65-80)	Connected Application of good prototypical standard. Used to evaluate overall suitability for a full project.	Lightweight messaging. Network/API programming. Architecture/ IOT Framework that mediates between devices and components.	>2 strands as above including more advanced knowledge and concepts.	Excellent Github Repo: Additional communication resources (e.g. instruction video, learning resources, Excellent ReadMe/installation guide)
Release 4 Outstanding (80-100)	Novel solution of clear applicability to specific domain. Could result in employment/internship offer.	All previous to excellent level. Excellent Use of Cloud/IoT specific platforms	All above, including self-acquired knowledge over and above module content.	All the above to excellent level, accessible project platform (e.g. web site, Social Media)



What Releases might look like:

Release 1(Core)

- At least one sensor or simulated data source (e.g. temperature, button, motion, External API).
- Regular data collection.
- Some local processing of raw values (e.g. thresholds, states, simple rules).
- One running program that shows behaviour clearly (console output acceptable).
- Basic logging/display (print values, write to file, simple terminal UI).
- Clear project description: problem, aim, and what you're building.

“My RPi reads sensor values every few seconds, checks if it's above a threshold, logs it, and shows a warning on screen.”

What Releases might look like:

Release 2(Good)

- **Two distinct components/processes** (e.g. sensor/edge node + service/dashboard).
- **At least one network connection** (MQTT, API, TCP/UDP, HTTP).
- **Structured data messages** (JSON recommended).
- **Meaningful data handling** beyond raw logging (averages, states, or simple analytics).
- **Basic dashboard or visualisation** (web page, terminal UI, or simple graph).
- **Initial architecture diagram** (boxes + arrows is fine).

“Edge device sends JSON readings to a second service via MQTT/HTTP. The service logs the messages and shows a simple dashboard page.”

What Releases might look like:

Release 3(Excellent)

- **Multi-node or multi-service architecture** (device → service → UI).
- **Clear message schema / API contract** documented in README.
- **Historical data handling** (DB/CSV with queries, charting, trend indicators).
- **Live updates** (MQTT subscription, WebSocket updates, or auto-refresh UI).
- **Derived metrics or smart behaviour** (alerts, recommendations, rules engine).
- **Refined UI** showing both current and historical data.
- **Clear, well-explained architecture diagram** (separation of concerns).

“My device publishes readings to an MQTT broker. A backend service stores them and exposes API endpoints. A web dashboard graphs trends and shows alerts when readings exceed limits. . If <some event> occurs, application does <x>”

What Releases might look like:

Release 4(Outstanding “stretch” features)

- **Cloud or remote deployment** (Azure IoT, running your broker/UI, or remote database).
- **Multiple communication methods** (e.g. MQTT + REST, or device → broker → cloud).
- **Configurable system** (thresholds, settings editable via UI or config file).
- **Robustness features** (reconnect logic, message validation, retry/backoff).
- **Advanced UI** (charts, filters, zones, device status, colour-coded alerts).
- **Self-learned technology** beyond module basics (e.g. advanced programming, containerisation, cloud services/DevOps etc.).
- **Insightful reflection** in video and readme : trade-offs, architecture choices, limitations.

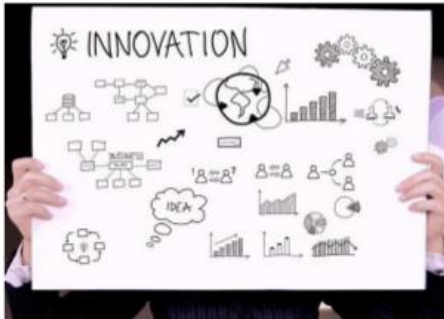
“My sensor node publishes to an MQTT broker on a cloud VM. A backend API ingests data into a cloud DB. A deployed dashboard visualises long-term trends, alerts, and device status. Thresholds can be changed from the web UI.”

Sample Project Ideas

- **Smart home/Monitoring device:**
 - Access and control a home device
 - Monitor/analyse home environment (medical/care apps)
 - Smart Plant Monitor
- **Who's there?:**
 - Detect and monitor devices on the home network (enhance lab work with cloud platform, extend to general smart home solution, use camera, use accelerometer/Gyro...)
- **Wether Weather 2.0?**
 - Extend/modify Web App assignment from last semester to connect/use your own environment device.
 - Already getting current/forecasted weather conditions. Combine with home environment data to analyse and/or control stuff (e.g. rain warning – take in the washing! Clear/Windy – hang out the washing!).



Other Considerations



- "Permissionless Innovation"!
 - You don't have to limit yourself to SenseHAT/RPi or module technologies. You can integrate other devices/ systems
 - We used Python but you can use any other programming languages/ frameworks if you want. (Remember, the RPi can run Node/Java/JavaScript)
- Have a look at other IoT project examples for inspiration.
- It's OK to "simulate" sensors
 - Use Packet Tracer
 - We realise you don't have a lab at your disposal. You can simulate data (e.g. GPS location.)